

torch.linalg.eigvalsh#

<https://pytorch.org/docs/stable/generated/torch.linalg.eigvalsh.html>

Description:

Computes the eigenvalues of a complex Hermitian or real symmetric matrix. Letting $K \in \mathbb{R} \text{ or } \mathbb{C}$, the eigenvalues of a complex Hermitian or real symmetric matrix $A \in K^{n \times n}$ are defined as the roots (counted with multiplicity) of the polynomial p of degree n given by $p(\lambda) = \det(A - \lambda I_n)$ where I_n is the n -dimensional identity matrix. The eigenvalues of a real symmetric or complex Hermitian matrix are always real. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. The eigenvalues are returned in ascending order.

Function Signature

```
torch.linalg.eigvalsh(A, UPL0='L', *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Returns:

A real-valued tensor containing the eigenvalues even when A is complex. The eigenvalues are returned in ascending order.

Code Examples

```
>>> A = torch.randn(2, 2, dtype=torch.complex128)
>>> A = A + A.T.conj() # creates a Hermitian matrix
>>> A
tensor([[2.9228+0.0000j, 0.2029-0.0862j],
        [0.2029+0.0862j, 0.3464+0.0000j]],
        dtype=torch.complex128)
>>> torch.linalg.eigvalsh(A)
tensor([0.3277, 2.9415], dtype=torch.float64)

>>> A = torch.randn(3, 2, 2, dtype=torch.float64)
>>> A = A + A.mT # creates a batch of symmetric matrices
>>> torch.linalg.eigvalsh(A)
tensor([[ 2.5797,  3.4629],
        [-4.1605,  1.3780],
        [-3.1113,  2.7381]], dtype=torch.float64)
```

Notes & Warnings

NOTE

Note When inputs are on a CUDA device, this function synchronizes that device with the CPU.

NOTE

See also `torch.linalg.eigh()` computes the full eigenvalue decomposition.

Detailed Documentation

Documentation

Computes the eigenvalues of a complex Hermitian or real symmetric matrix.

Letting K be $\mathbb{R}$ or $\mathbb{C}$, the eigenvalues of a complex Hermitian or real symmetric matrix $A \in K^{n \times n}$ are defined as the roots (counted with multiplicity) of the polynomial p of degree n given by

where I_n is the n -dimensional identity matrix. The eigenvalues of a real symmetric or complex Hermitian matrix are always real.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

The eigenvalues are returned in ascending order.

A is assumed to be Hermitian (resp. symmetric), but this is not checked internally, instead:

If `UPLO= 'L'` (default), only the lower triangular part of the matrix is used in the computation.

If `UPLO= 'U'`, only the upper triangular part of the matrix is used.

When inputs are on a CUDA device, this function synchronizes that device with the CPU.

`torch.linalg.eigh()` computes the full eigenvalue decomposition.

A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions consisting of symmetric or Hermitian matrices.

UPLO ('L', 'U', optional) – controls whether to use the upper or lower triangular part of A in the computations. Default: 'L'.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

A real-valued tensor containing the eigenvalues even when A is complex. The eigenvalues are returned in ascending order.